Lead 5b: the D_4-symmetrised S_5 as four weighted unknowns

2026-10-15, lane d4-s5. Study `research/d4_symmetrised_s5_20261015`. Protocol committed in `87fe78b63` before any measured run. Runs 06:38Z–07:43Z, one thread each, under a 2.5 GB cap, on a shared 4-vCPU container. Class: engineering, a constant of one phase. No end-to-end S.

The lead was already half-run

The encyclopedia lists the D_4-symmetrised S_5 as "never run". That entry is stale. Candidate C8 (PR #1785, `research/gaudry_quartic_d4_20261013`) ran it on 2026-10-13:

- form: five **unweighted** unknowns $(\sigma_1, \sigma_2, \sigma_3, \sigma_4, \pi)$ with $\pi^2 = \sigma_4$, Macaulay degree 14;
- matrix: 13412×11628;
- cost: $C_4 = 2.54 \cdot 10^{10}$, a factor of 48 below the plain S_4 form's $1.213 \cdot 10^{12}$. The projection was ≈ 630.

This study builds the **weighted** Macaulay solve that the GLV note §9.4 named and nobody built.

Construction (derived)

Curve and base. The curve is over $\mathbb{F}_{p^4}$ with $T = (x_0, 0)$ and $f'(x_0) = c^2$. The coordinate is $Y = \frac{x - x_0 - c}{x - x_0 + c}$, with $Y(P + T) = -Y(P)$. The base is $\{P : Y(P) \in \mathbb{F}_p\}$.

Group and invariants. $D_4 = (\mathbb{Z}/2)^3 \rtimes S_4$, of order 192, acts by permutations and by even sign changes, i.e. by even T-translates. The invariants are $\sigma_k = e_k(Y^2)$ and $\pi = \prod Y_i$.

System. Four Weil components in $(\sigma_1, \sigma_2, \sigma_3, \pi)$, each of weighted degree 8 under the weights $(2, 2, 2, 1)$.

Bézout. Two derivations agree:
- weighted: $8^4/(2 \cdot 2 \cdot 2 \cdot 1) = 512$;
- orbits: $4096/(192/24) = 512$.

Macaulay bound. $32 - 7 + 1 = 26$, at which the matrix has **4,200 columns**. C8 has 11,628 and the plain form 40,920.

Controls (measured)

- **Invariance detector.** A wrong c makes the invariant interpolation fail; the true c passes. Checked at $p = 13$ and $p = 269$.
- **Generic planted weighted systems.** 9 of 9 have quotient 512 and recover the planted root, at 1.39–$1.45 \cdot 10^9$.
- **Planted targets.** The planted quadruple is recovered by arm w in 41 of 42 and by C8 in 23 of 24.

Results (measured)

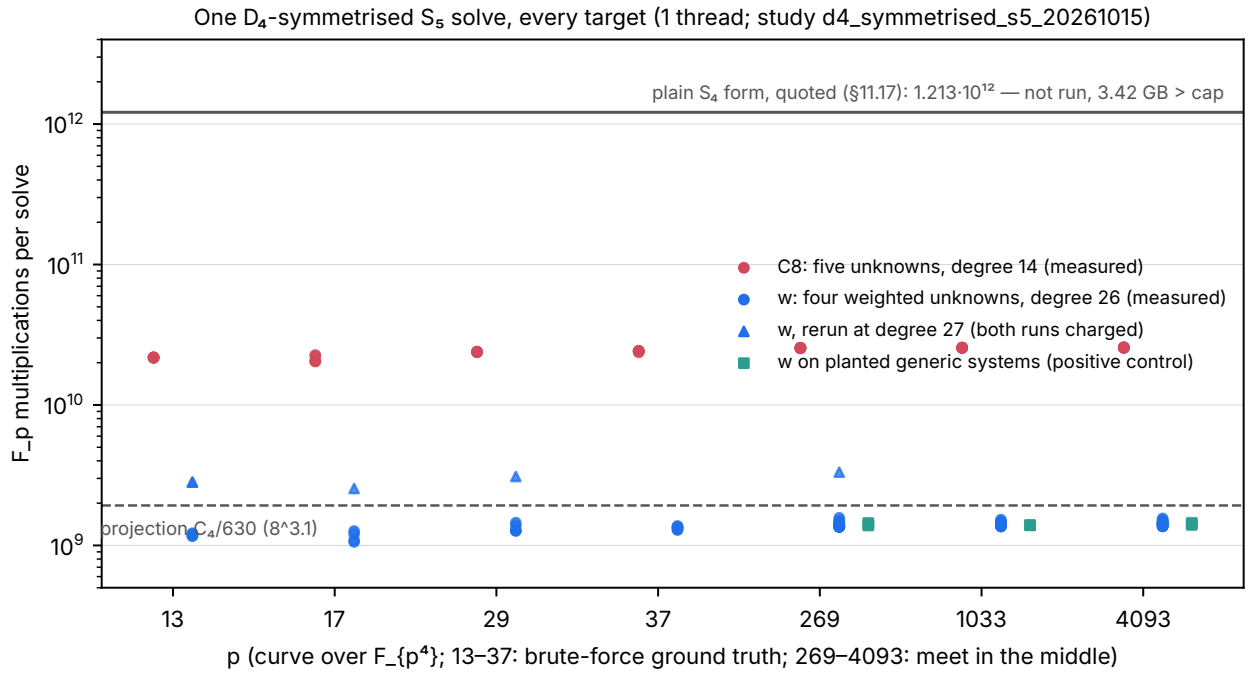


Figure 1: F_p multiplications per solve, every target. C8 (red) and the weighted form (blue) run on the same targets. The plain level is quoted from §11.17; it was not run, because its 3.42 GB matrix is over the cap.

p	C_w median	C_{c8} median	paired C_{c8}/C_w	$1.213 \cdot 10^{12}/C_w$	ground truth
13–37	1.21–$1.34 \cdot 10^9$	2.05–$2.42 \cdot 10^{10}$	17.9–18.5	905–1005	brute 23/24, MITM 23/24
269	$1.411 \cdot 10^9$	$2.542 \cdot 10^{10}$	18.0 (16.3–18.7)	860 (772–893)	MITM 20/20
1033	$1.406 \cdot 10^9$	$2.554 \cdot 10^{10}$	18.2 (17.4–18.7)	863 (823–888)	MITM 20/20
4093	$1.437 \cdot 10^9$	$2.561 \cdot 10^{10}$	17.8 (16.5–18.6)	844 (778–885)	MITM 20/20

- **Lift-back.** The split, F_p square roots, lifts and group checks of all 8 translates cost at most $2.8 \cdot 10^5$, which is at most $2 \cdot 10^{-4}$ of C_w.
- **Degree-27 reruns.** 5 of 101 solved targets had a degenerate eigenspace and reran at degree 27, charged at 2.5–$3.3 \cdot 10^9$ each.
- **One target unsolved.** At $p = 17$, planted target 4 has quotient 514 and is unsolved by both arms; brute force finds 4 decompositions there.
- **Exactness.** On every other target the two arms and the ground truth agree exactly.

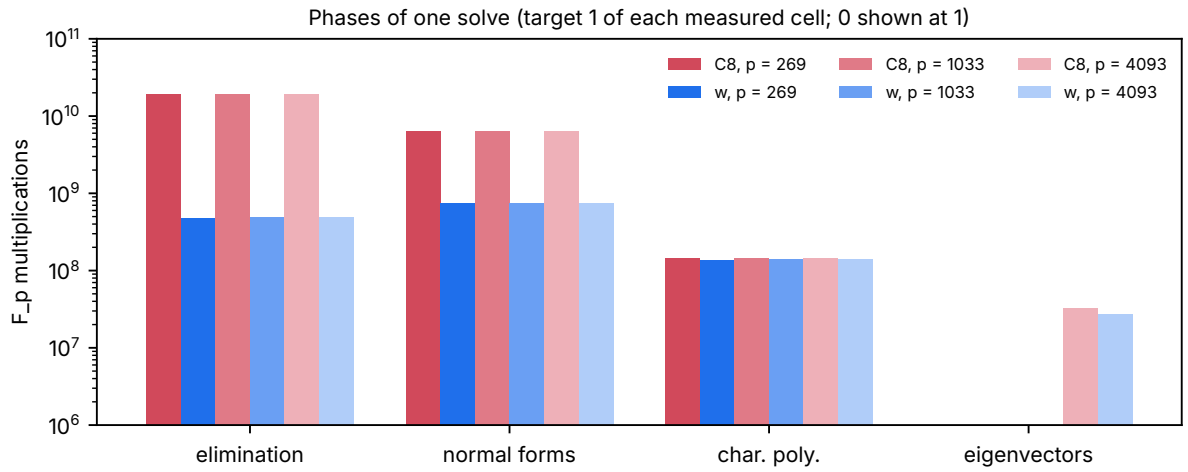


Figure 2: Phases of one solve (target 1, seed 1). Elimination falls by a factor of 40 and normal forms by 8. The characteristic polynomial is unchanged, since $D = 512$ in both arms.

Verdict under the registered rule

- **W1 and W2.**
  - W1 held at $p \geq 29$. At $p = 17$ the dimensions were 510–514.
  - W2 is **not met as registered**: one of 102 targets was unsolved. It was a flagged failure shared with C8, not a wrong set. The decision below therefore carries that caveat.
- **W3** held: the medians are flat in p to 2.2 %.
- **Lead 5b: "projection met".**
  - The median ratio to the plain C_4 is 844–863 at all three sizes, inside $[315, 1260]$.
  - It is not "accounting", because the lift-back is negligible.
- **Against C8: "advances on C8".** The paired ratio is at least 7.7 on all 47 paired targets.

The gap between 48 and 630 was the presentation, not the symmetry.

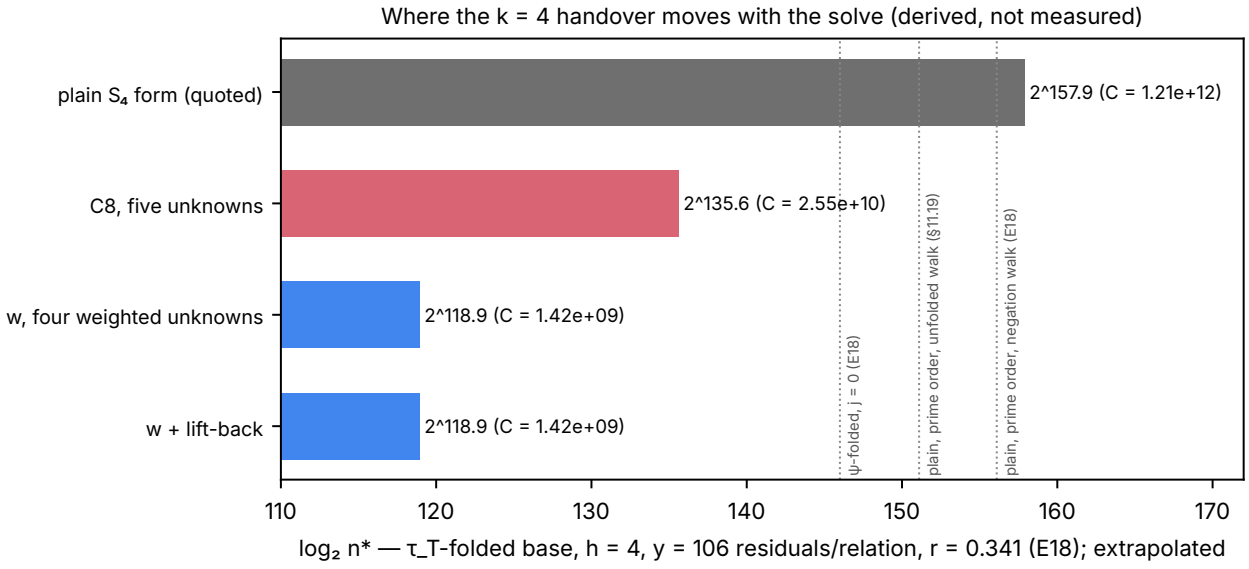


Figure 3: The $k = 4$ handover in E18's τ_T frame. Inputs: $h = 4$, 106 residuals per relation, $r = 0.341$ against the negation walk. The formula is $n^* = 157.9 - 4\log_2(1.213 \cdot 10^{12}/C)$. Derived and extrapolated, not measured.

Handover (derived, extrapolation)

On 2-torsion curves the weighted solve moves the τ_T handover from $2^{157.9}$ to $2^{118.9}$. C8 alone moves it to $2^{135.6}$.

Past the handover the method tends to 0.341 × the negation walk, and r_∞ is unchanged. The figure rests on the exponents $n^{1/4}$ and $n^{1/2}$, and on a C that is flat at $p \leq 4093$. No $k = 4$ run below rho exists.

Limits

- The plain arm was not paired on the same targets; it is over the cap.
- No relation phase or linear algebra was run.
- The degenerate targets' cause is unknown.
- The module's unit tests could not compile in the sparse checkout. The same checks ran as controls.
- No identity certificate was produced.